

```

Use this as the running checklist for the current ProofPro release candidate. Record failures with the
proof || Topic ID | 791402d5ac0fe1 || MQTT root | smartpidM5/proofpro/791402d5ac0fe1/ || U
ProofPro traffic: `` bash mosquitto_sub -h 10.0.1.203 -u proof -P proof \ -t 'smartpidM5/
live state against the ProofPro schema v2 contract. Add --conflict-test to also verify
'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"status":true}' ` Send heartbeat
proof -P proof \ -t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"workflow":
'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"workflow":"distillation","strat
` bash mosquitto_pub -h 10.0.1.203 -u proof -P proof \ -t 'smartpidM5/proofpro/791402c
ProofPro / DSPR400 Nomenclature In this protocol, dcx means dc1 or dc2, and rlx me
| Accel mode | Device-level, not per channel | | Acceleration end temp | distillation.ac
| distillation.run_power | Distill/run power | Element DC output percent after ACCEL cor
element | Direct/manual output; excluded from automatic program power | | Timer start
| distillation.finish_temp | dFSP | One threshold, not probe-independent values | | Fin
end, continue, shutoff if supported by UI | | Disabled relay | config.relays.rlx.mode:"
| May be operator-disengaged | | Proof relay | config.relays.rlx.mode:"remote_other" | R
in ms | | Cycle total time | rlx_cycle_ms | Cycle period | Full ON+OFF cycle in ms |
ended | program_ended.reason | END condition | Reasons: finish_timer, finish_temp, fi
starting near 123F. Use increasing targets so each run exercises more than one featu
RL2 remote_other; no heat required | Set relay/output mode to remote/manual test equ:
acceleration_power:100, run_power:35, timer_start_temp:140, timer_s:120, finish_action:
AccElement relay turns off at dAST; timer starts | | C. Timer END | 145-165F | distil
dtSP=155F; Timer 1:00; END on timer | program_ended.reason:"finish_timer"; DC1/DC2 0;
expected | Probe1 alone can trigger program_ended.reason:"finish_temp" | | E. Finish
program_ended.reason:"finish_temp" | | F. Cycle relay | Any stable temp | RL1 or RL2
Watchdog | Any safe state | Remote enabled, watchdog_s:30; stop Proof heartbeat | W
Resume ON | Device publishes controller_rebooted; Proof re-sends program and workflow
{"acceleration_enabled":true,"acceleration_end_temp":140,"acceleration_power":100,"ru
` bash mosquitto_pub -h 10.0.1.203 -u proof -P proof \ -t 'smartpidM5/proofpro/791402c
'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"distillation":
{"acceleration_enabled":true,"acceleration_end_temp":155,"acceleration_power":100,"ru
` `` bash mosquitto_pub -h 10.0.1.203 -u proof -P proof \ -t 'smartpidM5/proofpro/7914
'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"distillation":{"acceleration_en
'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"workflow":"distillation","strat
{"acceleration_enabled":false,"timer_s":300,"finish_temp":190,"finish_temp_probe":"pr
'{"workflow":"distillation","strategy":"program","action":"start"}' ` Cycle relay te
10.0.1.203 -u proof -P proof \ -t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m
-u proof -P proof \ -t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"r12_mode
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -m '{"r12":false}' ` Watchdog set
h 10.0.1.203 -u proof -P proof \ -t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \ -
'{"dc2_power":25}' ` ## 0. Safety and Setup - [ ] Confirm no hazardous load is conne
and events. - [ ] Open Proof and confirm the device appears online. - [ ] On the devi
publishes after reconnect. - [ ] Confirm retained status.schema version is 2. - [ ]

```

status.device_state, workflow, and strategy are present. - [] Confirm retained status
Confirm retained config.relays.rl1 and config.relays.rl2 are present. - [] Confirm live s
Proof settings, confirm relay modes and cycle timings come from retained config. Pass
State and Heartbeat - [] With Remote disabled, confirm remote_state:"OFF". - [] Try
- [] Confirm remote_state:"RDY" before Proof actively controls. - [] Send Proof hea
prevents output/program commands. - [] RDY means commandable. - [] ON means active
RL2. - [] Set retained relay mode to remote_other. - [] Confirm retained config.rela
remote_state:"ON". - [] Confirm live telemetry has state.relays.rlx.engaged:true. - []
off. - [] Confirm physical relay is forced OFF. - [] Change relay mode from off bac
from RDY and moves session to ON. - [] Changing mode always clears relay output befo
visibly different value than run power. - [] Set distillation.run_power to the expecte
programmed run. - [] Confirm status shows ACCEL. - [] Confirm DC tile primary val
until selected probe reaches distillation.acceleration_end_temp. - [] Confirm status ch
- [] Confirm only element DC outputs change to distillation.run_power. Pass criteria:
- [] Confirm programmed timer displays as HH:MM:SS. - [] Let timer expire with fi
Confirm Remote changes from ON to RDY. - [] Press Reset. - [] Confirm END clears
temp. - [] Repeat with distillation.finish_temp_probe set to probe2. - [] Confirm pro
Watchdog - [] Set Remote enabled. - [] Confirm watchdog is enabled. - [] Confirm \
[] Confirm DC1/DC2 drop to 0. - [] Confirm RL1/RL2 turn off. - [] Confirm watchdog_
Watchdog only arms during active Remote ON session. - [] Watchdog trip forces whole-
the controller. - [] Confirm retained status.auto_resume_enabled:true. - [] Confirm
{"workflow":"distillation","strategy":"program","action":"start"} and intended power/relay stat
Resume is discoverable from MQTT. - [] Proof does not rely on delayed firmware Auto
in Proof audit history. ## 7. Cycle Relay Mode Run for RL1 and RL2. - [] Set relay m
disengaged. - [] Send the matching {"rl1":true} or {"rl2":true}. - [] Confirm state.relay
Confirm physical relay turns off and stays off. Pass criteria: - [] Cycle timing fol
- [] Confirm Proof writes an equipment audit record for the outbound start command w
origin:"operator" and trigger:"run_start". - [] Toggle a relay from Proof. - [] Confirm
event includes the firmware event payload and current run_id/device_id. - [] If firm
Proof MQTT command has durable provenance. - [] Operator commands, heartbeat command
Regression - [] DS18B20 reads reasonable room temperature. - [] K-Type reads reason
[] PT100 2-wire T1 terminal pairing is tested and documented. - [] NTC is tested if
9. Audio End Notification - [] Record DSPR400 run-ending beep with MacBook QuickTime
Confirm speaker plays notification without boot pop or persistent whine. - [] From E
values. - [] No onboard microphone dependency exists. - [] Test chirp is accepted e
remote relay flow passes. - [] ACCEL -> RUN -> END` regression passes. - [] Watchdog s
committed and tagged.

ProofPro Bench Test Protocol

Use this as the running checklist for the current ProofPro release candidate. Record failures with the exact screen state, MQTT payload, Proof screen state, DSPR400 setting, and physical output state when possible.

Bench Unit

Item	Value
Device IP	10.0.1.60
Broker IP	10.0.1.203
Broker credentials	proof / proof
Topic ID	791402d5ac0fe1
MQTT root	smartpidM5/proofpro/791402d5ac0fe1/
Update path	OTA only for normal testing
Starting water temp for this run plan	About 123F

Command Shortcuts

Use these exact commands from a terminal if Proof does not expose a specific test control yet.

Subscribe to all ProofPro traffic:

```
mosquitto_sub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/#' -v
```

Run the schema v2 MQTT smoke test after firmware is flashed and online:

```
python3 scripts/mqtt_v2_smoke_test.py
```

This verifies retained `status`, retained `config`, and live `state` against the ProofPro schema v2 contract. Add `--conflict-test` to also verify `command_error.reason:"conflicting_alias"` for a non-energizing conflicting alias command.

Request retained status/config refresh:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"status":true}'
```

Send heartbeat:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"heartbeat":true}'
```

Start programmed power run:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"program","action":"start"}
```

Enter manual distillation without safe/off stop:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"manual","action":"start"}
```

Stop run:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"action":"stop"}'
```

Reset END/watchdog state:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"action":"reset"}'
```

Speaker chirp check:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"chirp":true}'
```

ProofPro / DSPR400 Nomenclature

In this protocol, `dcx` means `dc1` or `dc2`, and `r1x` means `r11` or `r12`. Use the numbered key in actual MQTT payloads.

Concept	ProofPro MQTT / Proof UI	DSPR400-style name	Notes
Acceleration enabled	<code>distillation.acceleration_enabled</code>	Accel mode	Device-level, not per channel
Acceleration end temp	<code>distillation.acceleration_end_temp</code>	<code>dAST</code>	When reached, status should change <code>ACCEL</code> -> <code>RUN</code>
Acceleration output power	<code>distillation.acceleration_power</code>	<code>dOUT</code>	DC output percent during <code>ACCEL</code>
Run output power	<code>distillation.run_power</code>	Distill/run power	Element DC output percent after <code>ACCEL</code> completes
Main DC output	<code>config.dc_outputs.dcx.mode:"element"</code>	Element output	Receives acceleration power and run power
Auxiliary DC output	<code>config.dc_outputs.dcx.mode:"auxiliary"</code>	Auxiliary element	Direct/manual output; excluded from automatic program power
Timer start temperature	<code>distillation.timer_start_temp</code>	<code>dtSP</code>	Temperature that starts the finish timer
Finish timer length	<code>distillation.timer_s</code>	Timer	Device-level finish timer duration
Finish temperature	<code>distillation.finish_temp</code>	<code>dFSP</code>	One threshold, not probe-independent values
Finish temp probe	<code>distillation.finish_temp_probe</code>	Finish source	<code>probe1</code> or <code>probe2</code> ; only this probe can end by temp
Finish action	<code>distillation.finish_action</code>	<code>dEO</code> / finish behavior	Canonical values: <code>end</code> , <code>continue</code> , <code>shutoff</code> if supported by UI
Disabled relay	<code>config.relays.rlx.mode:"off"</code>	Off/Disabled	Forced off, not commandable
Manual relay	<code>config.relays.rlx.mode:"manual_on_off"</code>	On/Off	Local UI control only
Accel relay	<code>config.relays.rlx.mode:"acc_element"</code>	AccElement	May be operator-disengaged
Proof relay	<code>config.relays.rlx.mode:"remote_other"</code>	Remote/Other	Proof direct relay control
Cycle relay	<code>config.relays.rlx.mode:"cycle"</code>	Cycle/Reflux timer	Uses <code>on_ms</code> and <code>cycle_ms</code>
Cycle on time	<code>rlx_on_ms</code>	Cycle on time	Relay ON portion in ms
Cycle total time	<code>rlx_cycle_ms</code>	Cycle period	Full ON+OFF cycle in ms
Remote ready	<code>remote_state:"RDY"</code>	Remote ready	Commands accepted, watchdog not active yet
Remote active	<code>remote_state:"ON"</code>	Remote on	Proof is controlling; heartbeat expected
Program ended	<code>program_ended.reason</code>	END condition	Reasons: <code>finish_timer</code> , <code>finish_temp</code> , <code>finish</code>
Command provenance	Proof audit <code>origin</code> / <code>trigger</code>	Audit trail	Distinguishes operator, heartbeat, restore, system actions

Dynamic Heat-Up Test Matrix

These runs are designed for a boiler starting near 123F . Use increasing targets so each run exercises more than one feature before the water approaches boiling.

Run	Starting Temp	ProofPro Program Settings	DSPR400 Settings To Match	Primary Checks
A. Remote relay smoke test	Any	Relay RL2 <code>remote_other</code> ; no heat required	Set relay/output mode to remote/manual test equivalent	Proof can toggle relay from <code>RDY</code> ; telemetry mode matches retained config
B. Accel transition	123-145F	<code>distillation.acceleration_enabled:true , acceleration_end_temp:140 , acceleration_power:100 , run_power:35 , timer_start_temp:140 , timer_s:120 , finish_action:"continue"</code>	Accel enabled; <code>dAST=140F</code> ; <code>dOUT=100</code> ; run power 35% ; Timer start <code>dtSP=140F</code> ; Timer 2:00 ; finish continue	Status starts <code>ACCEL</code> , then changes to <code>RUN</code> ; DC power changes to run power; AccElement relay turns off at <code>dAST</code> ; timer starts
C. Timer END	145-165F	<code>distillation.acceleration_enabled:true , acceleration_end_temp:155 , acceleration_power:100 , run_power:35 , timer_start_temp:155 , timer_s:60 , finish_action:"end"</code>	<code>dAST=155F</code> ; <code>dOUT=100</code> ; run power 35% ; <code>dtSP=155F</code> ; Timer 1:00 ; END on timer	<code>program_ended.reason:"finish_timer"</code> ; DC1/DC2 0; RL1/RL2 off; Remote <code>ON -> RDY</code>
D. Finish temp probe1	165-185F	<code>finish_temp:175 , finish_temp_probe:"probe1" , timer_s:300 , finish_action:"end"</code>	<code>dFSP=175F</code> ; finish source probe1; timer longer than expected	Probe1 alone can trigger <code>program_ended.reason:"finish_temp"</code>
E. Finish temp probe2	175-200F	<code>finish_temp:190 , finish_temp_probe:"probe2" , timer_s:300 , finish_action:"end"</code>	<code>dFSP=190F</code> ; finish source probe2; timer longer than expected	Probe2 alone can trigger <code>program_ended.reason:"finish_temp"</code>
F. Cycle relay	Any stable temp	RL1 or RL2 mode:"cycle" , on_ms:1000 , cycle_ms:5000 , then <code>rlx:true</code>	Cycle/reflux relay; ON 1 sec, total 5 sec	<code>state.relays.rlx.engaged:true</code> ; physical <code>state.relays.rlx.state</code> blinks; local disengage stays off
G. Watchdog	Any safe state	Remote enabled, <code>watchdog_s:30</code> ; stop Proof heartbeat	Watchdog/remote safety if available	After >30s without heartbeat: DC1/DC2 0, RL1/RL2 off, <code>watchdog_safe</code> , Remote <code>RDY</code>
H. Reboot restore	Any safe state	Active Proof run, Auto Resume ON	Auto Resume ON	Device publishes <code>controller_rebooted</code> ; Proof re-sends program and workflow start command

Program Command Payloads

Run B, Accel transition:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"distillation":  
{ "acceleration_enabled":true,"acceleration_end_temp":140,"acceleration_power":100,"run_power":35,"timer_start_temp":140
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"program","action":"start"}'
```

Run C, Timer END:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"distillation":  
{"acceleration_enabled":true,"acceleration_end_temp":155,"acceleration_power":100,"run_power":35,"timer_start_temp":155
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"program","action":"start"}'
```

Run D, Finish temp from probe1:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"distillation":  
{"acceleration_enabled":false,"timer_s":300,"finish_temp":175,"finish_temp_probe":"probe1","finish_action":"end"}}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"program","action":"start"}'
```

Run E, Finish temp from probe2:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"distillation":  
{"acceleration_enabled":false,"timer_s":300,"finish_temp":190,"finish_temp_probe":"probe2","finish_action":"end"}}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"workflow":"distillation","strategy":"program","action":"start"}'
```

Cycle relay test, RL1:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r11_mode":"cycle","r11_on_ms":1000,"r11_cycle_ms":5000}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r11":true}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r11":false}'
```

Remote relay test, RL2:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r12_mode":"remote_other"}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r12":true}'
```

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \  
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \  
-m '{"r12":false}'
```

Watchdog setup:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \
-m '{"watchdog_enabled":true,"watchdog_s":30}'
```

DC output role setup:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \
-m '{"dc1_mode":"element","dc2_mode":"auxiliary}"'
```

Auxiliary DC direct command:

```
mosquitto_pub -h 10.0.1.203 -u proof -P proof \
-t 'smartpidM5/proofpro/791402d5ac0fe1/commands' \
-m '{"dc2_power":25}'
```

0. Safety and Setup

- [] Confirm no hazardous load is connected to DC1 during bench firmware work.
- [] Confirm OTA is used for firmware updates.
- [] Do not USB-flash with a heater or hazardous load connected to DC1.
- [] Subscribe to status, config, state, and events.
- [] Open Proof and confirm the device appears online.
- [] On the device, start from the Power screen.
- [] Confirm Remote is enabled on the device before testing Proof commands.

1. Retained Status and Config

- [] Reboot the device.
- [] Confirm retained `status` publishes after reconnect.
- [] Confirm retained `status.schema_version` is `2`.
- [] Confirm retained `status.firmware_version` is `0.3.0`.
- [] Confirm retained `status.unit` is `F` or `C`.
- [] Confirm retained `status.remote_state` is `OFF`, `RDY`, or `ON`.
- [] Confirm retained `status.device_state`, `workflow`, and `strategy` are present.
- [] Confirm retained `status.watchdog_enabled` and `watchdog_s` are present.
- [] Confirm retained `config.distillation` is present.
- [] Confirm retained `config.dc_outputs.dc1.mode` and `config.dc_outputs.dc2.mode` are present.
- [] Confirm retained `config.relays.r11` and `config.relays.r12` are present.
- [] Confirm live `state` publishes while the device is idle.
- [] In Proof onboarding/settings, confirm unit is auto-filled from status.
- [] In Proof settings, confirm program defaults come from retained config.
- [] In Proof settings, confirm relay modes and cycle timings come from retained config.

Pass criteria:

- [] Proof can read retained status/config without requiring manual unit or watchdog entry.
- [] Retained config and live telemetry do not disagree on relay mode after reboot.

2. Remote State and Heartbeat

- [] With Remote disabled, confirm `remote_state:"OFF"`.
- [] Try a Proof output/program command with Remote OFF.
- [] Confirm command is rejected or ignored and no output energizes.
- [] Enable Remote on the device.
- [] Confirm retained/live `remote_enabled:true`.
- [] Confirm `remote_state:"RDY"` before Proof actively controls.
- [] Send Proof heartbeat or a remote command.

- [] Confirm `remote_state:"ON"` .
- [] Stop Proof control or disable Remote.
- [] Confirm Remote returns to `RDY` or `OFF` as expected.

Pass criteria:

- [] Remote OFF prevents output/program commands.
- [] RDY means commandable.
- [] ON means active Proof session and watchdog-protected.
- [] Proof audit logs heartbeat commands with `origin:"heartbeat"` and `trigger:"heartbeat_tick"`, not as operator commands.

3. Proof Remote Relay Flow

Run for RL1 and RL2.

- [] Set retained relay mode to `remote_other` .
- [] Confirm retained `config.relays.rlx.mode:"remote_other"` .
- [] Confirm live `state.relays.rlx.mode:"remote_other"` .
- [] Confirm Remote is `RDY` .
- [] Send the matching `{"r11":true}` or `{"r12":true}` from Proof.
- [] Confirm `remote_state:"ON"` .
- [] Confirm live telemetry has `state.relays.rlx.engaged:true` .
- [] Confirm physical relay output is ON.
- [] Send the matching `{"r11":false}` or `{"r12":false}` .
- [] Confirm `state.relays.rlx.engaged:false` .
- [] Confirm physical relay output is OFF.
- [] Change relay mode to `off` .
- [] Confirm physical relay is forced OFF.
- [] Change relay mode from `off` back to `remote_other` .
- [] Confirm mode change is accepted and live telemetry updates.

Pass criteria:

- [] Proof does not need to resend relay mode just to fix stale live state.
- [] Relay command works from RDY and moves session to ON.
- [] Changing mode always clears relay output before the new mode is used.

4. Accel to Run

- [] Configure acceleration enabled.
- [] Set `distillation.acceleration_end_temp` below a reachable test temperature.
- [] Set `distillation.acceleration_power` to a visibly different value than run power.
- [] Set `distillation.run_power` to the expected RUN output percent.
- [] If a DC output is auxiliary, confirm it stays off during ACCEL/RUN unless directly commanded.
- [] Set at least one relay to `acc_element` if testing AccElement.
- [] Start programmed run.
- [] Confirm status shows `ACCEL` .
- [] Confirm DC tile primary value shows actual accel power.
- [] Confirm queued post-accel run power appears separately as small secondary text.
- [] Confirm AccElement relay engages while accel is active.
- [] Heat or simulate until selected probe reaches `distillation.acceleration_end_temp` .
- [] Confirm status changes to `RUN` .
- [] Confirm queued run power becomes the primary actual power value after acceleration ends.
- [] Confirm AccElement relay turns off.
- [] Confirm telemetry publishes `accel_complete` .
- [] Confirm only `element` DC outputs change to `distillation.run_power` .

Pass criteria:

- [] Accel completes as a device-level phase.
- [] No channel remains stuck in `ACCEL` after threshold is reached.

5. Timer and END

- [] Set programmed timer duration.
- [] Start a programmed run.
- [] Confirm programmed timer displays as `HH:MM:SS`.
- [] Let timer expire with finish action `end`.
- [] Confirm status changes to `END`.
- [] Confirm DC1/DC2 go 0.
- [] Confirm RL1/RL2 are off.
- [] Confirm `events/standard` publishes `program_ended` with `reason:"finish_timer"`.
- [] Confirm Remote changes from `ON` to `RDY`.
- [] Press Reset.
- [] Confirm `END` clears and the program does not start by itself.

Finish temperature:

- [] Set one device-level `finish_temp`.
- [] Set `distillation.finish_temp_probe` to `probe1`.
- [] Confirm only `probe1` can trigger finish-by-temp.
- [] Repeat with `distillation.finish_temp_probe` set to `probe2`.
- [] Confirm `program_ended.reason:"finish_temp"`.

Pass criteria:

- [] Timer `END` and finish-temp `END` both produce device-level `program_ended`.
- [] `END` safe/off behavior applies to both DC outputs and both relays.

6. Watchdog

- [] Set Remote enabled.
- [] Confirm watchdog is enabled.
- [] Confirm `watchdog_s` is between 30 and 60.
- [] Start a remote-controlled run.
- [] Confirm Proof sends heartbeat every 10 seconds.
- [] Stop heartbeats/remote commands.
- [] Wait longer than `watchdog_s`.
- [] Confirm DC1/DC2 drop to 0.
- [] Confirm RL1/RL2 turn off.
- [] Confirm `watchdog_safe` event publishes with `watchdog_s`.
- [] Confirm Remote returns from `ON` to `RDY`.
- [] Send a valid command or heartbeat.
- [] Confirm `watchdog_cleared` event publishes.

Pass criteria:

- [] Watchdog only arms during active Remote `ON` session.
- [] Watchdog trip forces whole-device safe/off.
- [] Watchdog can be cleared by later accepted remote traffic.

6a. Controller Reboot and Proof Restore

- [] Set Auto Resume `ON` locally.
- [] Start an active Proof run.
- [] Reboot the controller.
- [] Confirm retained `status.auto_resume_enabled:true`.
- [] Confirm `events/standard` publishes `controller_rebooted` with `reset_reason`.
- [] Confirm Proof detects the reboot/reconnect.
- [] Confirm Proof re-sends the active program payload.
- [] Confirm Proof re-sends `{"workflow":"distillation","strategy":"program","action":"start"}` and intended power/relay state.
- [] Confirm Proof audit logs restore commands with `origin:"restore"` and `trigger:"reconnect_restore"`.
- [] Confirm the controller returns to the intended active run state.

Pass criteria:

- [] Auto Resume is discoverable from MQTT.
- [] Proof does not rely on delayed firmware Auto Resume as the only restore path.
- [] Controller reboot does not leave the power controller idle during an active Proof run.
- [] Reconnect restore is distinguishable from a manual/operator run start in Proof audit history.

7. Cycle Relay Mode

Run for RL1 and RL2.

- [] Set relay mode to `cycle`.
- [] Set `on_ms`.
- [] Set `cycle_ms`.
- [] Confirm retained config reports mode and timing.
- [] Confirm live telemetry reports `state.relays.rlx.mode:"cycle"`.
- [] Confirm relay starts disengaged.
- [] Send the matching `{"r11":true}` or `{"r12":true}`.
- [] Confirm `state.relays.rlx.engaged:true`.
- [] Confirm physical relay blinks/cycles.
- [] Confirm telemetry `state.relays.rlx.state` follows actual GPIO ON/OFF state.
- [] Send the matching `{"r11":false}` or `{"r12":false}`.
- [] Confirm physical relay turns off and stays off.

Pass criteria:

- [] Cycle timing follows configured `on_ms` / `cycle_ms`.
- [] `state.relays.rlx.engaged` means armed; `state.relays.rlx.state` means actual physical state.

7a. Proof Audit and END Decisions

- [] Start a ProofPro run from Proof.
- [] Confirm Proof writes an equipment audit record for the outbound start command with timestamp, run_id, device_id, topic, payload, origin, and trigger.
- [] Confirm the run also has a `mqtt_command_sent` event for the outbound start command.
- [] Confirm run start uses `origin:"operator"` and `trigger:"run_start"`.
- [] Toggle a relay from Proof.
- [] Confirm relay command audit uses `origin:"operator"` and `trigger:"relay_toggle_button"`.
- [] Let firmware publish `program_ended`.
- [] Confirm Proof writes `proofpro_program_end_decision`.
- [] Confirm the decision event includes the firmware event payload and current run_id/device_id.
- [] If firmware event includes optional `source`, confirm Proof preserves it as `firmware_source`.
- [] If firmware event omits `source`, confirm Proof handles it without error.

Pass criteria:

- [] Every outbound Proof MQTT command has durable provenance.
- [] Operator commands, heartbeat commands, reconnect restore, and settings sync can be distinguished in Proof audit history.
- [] Firmware END handling records Proof's next decision instead of only the incoming firmware event.

8. Probe Regression

- [] DS18B20 reads reasonable room temperature.
- [] K-Type reads reasonable temperature against reference.
- [] PT100 3-wire T1 reads with documented offset.
- [] PT100 3-wire T2 reads with documented offset.
- [] PT100 2-wire T2 works with documented terminal pairing.
- [] PT100 2-wire T1 terminal pairing is tested and documented.
- [] NTC is tested if a probe is available; otherwise leave unvalidated.

Pass criteria:

- [] Unsupported/unvalidated probe routes are documented clearly.

- [] Invalid probe states display/publish `ERR` / `temp_valid:false`.

9. Audio End Notification

- [] Record DSPR400 run-ending beep with MacBook QuickTime.
- [] Save recording under `docs/audio/` or another project path.
- [] Extract beep frequencies.
- [] Extract beep durations and gaps.
- [] Implement synthesized tone sequence in firmware.
- [] Confirm speaker plays notification without boot pop or persistent whine.
- [] From END state, send `{"chirp":true}` or `{"audio":"chirp"}`.
- [] Confirm `audio_chirp` event publishes and the test chirp plays.

Pass criteria:

- [] End notification is generated from tone frequency/duration values.
- [] No onboard microphone dependency exists.
- [] Test chirp is accepted even when Remote is `RDY` and program state is `END`.

Release Decision

Do not call the firmware shippable until these are checked:

- [] Proof retained status/config onboarding passes.
- [] Proof remote relay flow passes.
- [] `ACCEL -> RUN -> END` regression passes.
- [] Watchdog safe/off regression passes.
- [] Cycle relay regression passes or is explicitly deferred.
- [] PT100 2-wire CH1 is confirmed or explicitly unsupported.
- [] USB flashing hazard is documented in release notes.
- [] Current tree is committed and tagged.